

FR-016 — Multi-Agent Orchestration

FR016 — `sources/acceptance.md`

Masterclass source package snapshot

Functional Requirement	FR-016
Edition	Engineering Learning Academy — Source Package snapshot
Status	**Complete / Frozen / Accepted
Audience	Academy generation and study (snapshot; SoT remains authoritative)
Document type	Source package snapshot (PDF)

Faithful rendering of a regenerable Masterclass source snapshot. Do not hand-edit; regenerate via python scripts/build_masterclass_package.py. Repository documentation remains the source of truth.

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FR-016 — Multi-Agent Orchestration

Status: Complete / Frozen / Accepted

Date: 2026-08-06

Documentation close-out: 2026-08-06

Recommendation: ACCEPT AND FREEZE — LEARNING PROOF COMPLETE

Binding product posture: GO AS LEARNING PROOF ONLY (M2; unchanged through M4)

Engineering Learning Academy: Ready — canonical engineering record = this report; attachable package = [masterclass/FR016/](#) ([README.md](#) , [MANIFEST.md](#) , regenerable [sources/](#))

Next: Horizon 1B (FR-018+) on owner request — **not gated on FR-017. FR-017 is complete and frozen** ([fr017_agent_evaluation_observability.md](#)). Ordinary prep remains `cic agent run`.

ADR: [ADR-008](#) (Accepted — M1–M4 close-out)

Milestones (historical records): [M0](#) (Accepted with revisions), [M1](#), [M2](#) (**GO AS LEARNING PROOF ONLY**), [M3](#), [M4](#).

This document is the **canonical engineering record** for FR-016. Milestone reports remain historical; do not reopen exit criteria without owner request.

1. Executive Summary

FR-016 delivers a **constrained multi-agent learning proof**:

Role	Authority
DOS	Deterministic Orchestration Supervisor — delegates only ; no domain work
BOPA	Frozen FR-015 preparation specialist — mutating allow-list unchanged
OBS	Read-only Operational Briefing Specialist — compose briefs; never mutate

Typed handoffs, DelegationPolicy, per-specialist ToolPolicy, append-only parent/child audit, checkpoint/resume, and loop controls are implemented and evaluated.

Near-term product value is modest. Direct `cic agent run` remains the preferred daily preparation path. FR-016 is optional teaching/substrate tooling — not the default workflow.

Milestone	Delivered
M0	Topology spike; Prep/Truth/Review theatre rejected
M1	Contracts + ADR-008
M2	DOS + OBS + BOPA adapter; corpus; GO AS LEARNING PROOF ONLY
M3	<code>cic agent orchestrate</code> owner CLI
M4	Final corpus 20/20, safety/product review, study-aid source, docs freeze

Package: `career_intelligence.multi_agent`

CLI: `cic agent orchestrate {run,resume,show,history,list,check-delegation}`

Goals: `brief` | `prepare` | `prepare_then_brief`

2. Business Problem

After FR-015, the owner can prepare a single Opportunity for review with BOPA. Remaining questions were product and engineering honesty questions, not missing prepare/truth tools:

1. When does a **second agent** add value beyond one bounded agent?
2. How do we teach **permission-separated multi-agent** design without shipping theatre?
3. How do we keep Horizon 1 job-acquisition first while capturing interview-transferable multi-agent engineering skill?

FR-016 exists to answer those questions with evidence — not to replace daily prep UX.

3. Engineering Problem

How can CIC introduce multi-agent orchestration that:

1. Separates supervisor authority from specialist authority?
2. Uses typed, auditable handoffs instead of free-form agent chat?
3. Keeps BOPA's ToolPolicy and allow-list frozen?
4. Adds a second specialist only where the permission boundary is genuinely different?
5. Fails closed on loops, injection, truth waiver, submission, and pipeline mutation?
6. Remains evaluable offline without requiring an LLM supervisor?

The answer is **not** Prep/Truth/Review personas, and **not** an LLM coordinator with shared tools. It is **DOS (delegate-only) + frozen BOPA + read-only OBS**.

4. Why FR-015 alone was insufficient (for this milestone)

FR-015 (BOPA) is sufficient — and preferred — for **ordinary preparation**.

FR-015 alone was insufficient for the **FR-016 learning objective** because:

- It does not demonstrate supervisor/specialist separation.
- It does not provide a second specialist with a **distinct** ToolPolicy.
- It does not exercise typed cross-specialist handoffs or parent/child audit.
- It does not teach why multi-agent theatre fails.

FR-016 therefore complements BOPA as substrate and curriculum; it does **not** obsolete `cic agent run`.

5. Alternatives considered

Alternative	Outcome
Defer FR-016; keep BOPA only	Valid commercially; owner chose constrained learning path
Prep / Truth / Review specialist cast	Rejected — multi-agent theatre
Broaden BOPA to absorb briefing	Rejected — mixes brief-only with mutating prepare
LLM supervisor with chat handoffs	Rejected — weak control; injection surface
Hierarchical agent team	Rejected — unnecessary for single-user CIC
Framework graph (LangGraph / Agents SDK / CrewAI / MAF)	Deferred / Out of Scope — ADR-003 unmet
Productise as default daily workflow	Rejected — evidence showed modest value

6. Rejected architectures (multi-agent theatre)

```
Rejected:
  Supervisor
    ├── PrepAgent  → same prepare service
    ├── TruthAgent → same truth service
    └── ReviewAgent → same owner-review stop
```

Why rejected: no distinct ToolPolicy; personas rename work already owned by BOPA/services; weakens explainability; teaches the wrong interview lesson.

7. Final DOS / BOPA / OBS architecture



Domain SoTs remain authoritative. Orchestration audit is additive recovery data. Package is distinct from FR-008 `orchestration` and FR-015 `agent`.

8. ADR-008

[ADR-008](#) freezes:

1. Constrained multi-agent under `career_intelligence.multi_agent`
2. Topology: DOS + BOPA + OBS + typed handoffs
3. DelegationPolicy admits specialists; per-specialist ToolPolicy admits actions
4. Deterministic default; optional LLM never expands allow-lists
5. Explicit theatre rejection (Prep/Truth/Review personas)
6. Commercial honesty: learning/substrate; no strong near-term product claim
7. Out of scope: discovery, submit, pipeline mutation, recruiter contact, truth waiver, free-form chat, frameworks without new ADR

9. Runtime behaviour

DOS loop: observe SoT → select specialist → evaluate DelegationPolicy → create typed handoff → execute OBS or BOPA adapter → append audit → re-observe or stop.

Goal	Typical path
brief	DOS → OBS → briefing_complete
prepare	DOS → BOPA → owner stop
prepare_then_brief	DOS → BOPA → OBS → owner stop

Non-obvious routing: interviewing / ambiguity / truth blockers → OBS before (or instead of) BOPA.
Injection text in owner notes cannot alter delegation.

10. Authority boundaries

Actor	May	Must not
DOS	Observe, select, hand off, stop	Domain services, submit, pipeline, waive, inherit tools
BOPA	inspect, prepare, verify, validate truth, owner review, stop	submit, pipeline, discovery, waive, FR-008 repair
OBS	inspect readiness/pipeline/truth/history, compose brief, recommend	prepare, validate, submit, pipeline, waive
Handoff	Carry typed goal + hash + expected output	Grant tools or widen policy

11. ToolPolicy separation

Policy	Admits
DelegationPolicy	Whether DOS may invoke OBS or BOPA
BOPA ToolPolicy (<code>evaluate_action_policy</code>)	BOPA allow-listed actions only
OBS ToolPolicy (<code>evaluate_obs_action_policy</code>)	OBS read-only allow-list only

Global orchestration limits (max steps, max visits, repeated/circular delegation, no-progress) are separate from specialist ToolPolicies. No specialist inherits another's permissions.

12. Typed handoffs

Handoffs are append-only, policy-validated, and idempotent. Lifecycle: `pending` → `policy_blocked` | `accepted` → `executing` → `completed` | `stopped` (also `rejected` / `stale` / `cancelled`).

They carry: source, target, requested goal, observed state hash, expected output kind, policy decision, child AgentRun or OperationalBrief ref. They never carry free-form chat or tool grants.

13. Audit model

Parent `OrchestrationRun` under `data/orchestration_runs/` records events, handoff IDs, specialist visits, child AgentRun IDs, last brief ID, stop reason, and owner action. Child BOPA runs remain under `data/agent_runs/`. Audits are **not** a second Opportunity SoT.

Owner presentation reconstructs: `goal` → `observation` → `selection` → `policy` → `handoff` → `authority` → `child` → `stop` → `next action` (+ elapsed time).

14. Resume behaviour

Resume re-inspects authoritative state; rejects stale/contradictory checkpoints; avoids repeating completed BOPA work; avoids regenerating unchanged OBS briefs; requires `--approve`; preserves global and specialist limits.

15. Evaluation results

Final corpus `run_corpus()` : **20/20 PASS** ([M4](#)).

Covers brief, prepare, prepare_then_brief, interviewing, truth-blocked, material-benefit, illegal delegation, OBS mutate, DOS domain-work, repeated/ circular delegation, stale checkpoint, partial BOPA resume, unchanged OBS resume, provider outage, prompt injection, pipeline/submission/truth-waiver safety, and step/visit limits.

16. Manual validation

Offline A–H journeys PASS via `scripts/run_fr016_m4_manual.py`.

Live (safe): brief-only on interviewing opportunity `opp_01KY8RFAH81M9V30ZVH9TM09T5` → `DOS` → `OBS` → `briefing_complete`; pipeline note that preparation is usually unnecessary; **no domain mutation**.

Mutating live prepare remains optional OAT outside freeze criteria.

17. Product-value assessment

Question	Answer
Does DOS improve ordinary preparation?	No (materially)
Does OBS remove an interpretation task?	Yes (pipeline/truth briefs)
Does <code>prepare_then_brief</code> help?	Modest convenience
Permission separation worth complexity?	Yes for learning/substrate; not for daily prep
Ready as default daily workflow?	No — remain optional

18. Learning-value assessment

High. FR-016 is justified primarily as:

- production practice of permission-separated multi-agent design;
 - explicit rejection of multi-agent theatre;
 - interview-transferable artefacts (authority matrix, handoff lifecycle, Q&A);
 - substrate for a future specialist with a genuinely different boundary (e.g. Job Discovery).
-

19. Commercial assessment

Near-term commercial value: **modest**. Horizon 1 dual-value for daily applications is better served by direct BOPA. Commercial upside rises if/when Job Discovery or another second mutating authority appears — not by renaming BOPA roles.

Binding freeze posture: **GO AS LEARNING PROOF ONLY.**

20. Technical debt

Item	Class
Live LLM supervisor evaluation	Deferred → FR-017 / Future
Richer multi-specialist topology	Future FR
Scheduled briefings	Future FR / Out of Scope now
Job Discovery integration	Future FR
Orchestration replay / golden latency harness	Deferred → FR-017
Richer observability (token/cost/provider aggregates)	Deferred → FR-017
Framework migration	Out of Scope (ADR-003)
Visual study-aid / Masterclass generation	Deferred (source captured here)
Mutating live OAT dogfooding	Accepted (optional)

No debt implemented during close-out.

21. Risks

Risk	Mitigation / status
Owners treat orchestrate as default prep	Banner + docs: prefer <code>cic agent run</code>
Theatre creep (new personas)	ADR-008 rejection; freeze
Privilege escalation via handoff	Typed handoff; no tool grants
Scope leak into FR-017 / 1B	Explicit "do not auto-start"
Overclaiming product value	Learning-proof verdict frozen

22. Engineering retrospective

Theme	Record
What worked well	Delegate-only DOS; OBS as distinct read-only boundary; typed handoffs; honest go/no-go; BOPA left frozen
What did not	Multi-agent did not beat direct BOPA on ordinary prep latency/complexity
What surprised us	OBS value on interviewing/truth briefs without needing a second mutating specialist
What would be repeated	Mandatory go/no-go before productisation; theatre rejection early; deterministic default
What would be changed	Would not market FR-016 as daily UX; would wait for Job Discovery before richer topology
Why Learning Proof	Evidence showed substrate + learning value > near-term product gain
Why not preferred daily workflow	Extra observe/handoff steps; owner interpretation cost; direct BOPA already sufficient

23. Lessons learned (interview-ready)

1. **Authority boundaries** — supervisor $\neq$ specialist; specialist $\neq$ specialist.
 2. **Deterministic orchestration** — routing and admission stay deterministic.
 3. **Specialist permissions** — per-specialist ToolPolicy is mandatory.
 4. **Theatre rejected** — same tools + new names $\neq$ multi-agent value.
 5. **Typed handoffs matter** — audit, idempotency, fail-closed control.
 6. **DOS performs no domain work** — prevents supervisor super-user escalation.
 7. **OBS remains read-only** — briefing delta without widening BOPA.
 8. **Direct BOPA preferred** for ordinary preparation.
-

24. Operational readiness

Item	Status
Learning-proof CLI operable	Yes (<code>cic agent orchestrate</code>)
Daily prep path	Prefer <code>cic agent run</code>
Safety gates	Pass (corpus + manual)
Live mutating dogfood	Optional OAT (not freeze blocker)
Default owner workflow	Remains direct BOPA

25. Study-aid source material (no visuals generated)

25.1 Sixty-second explanation

Career Intelligence Copilot's FR-016 is a learning proof of safe multi-agent orchestration. A deterministic supervisor (DOS) never prepares packages or submits applications — it only chooses specialists. BOPA remains the existing bounded preparation agent. OBS is a read-only briefer for pipeline and truth context. Typed handoffs and policies keep authorities separate. For everyday prep, use `cic agent run`; use `cic agent orchestrate` to inspect and learn the delegation story.

25.2 Five-minute lesson outline

1. Problem: when does multi-agent help vs theatre?
2. Rejected design: Prep/Truth/Review personas.
3. Accepted topology: DOS + BOPA + OBS.
4. Authority matrix walkthrough.
5. Happy path vs truth-blocked sequences.
6. Why daily prep stays on direct BOPA.
7. Interview Q&A from §25.12–25.13.

25.3 Final architecture (Mermaid)

```
flowchart TD
  Owner["Owner CLI<br/>cic agent orchestrate"] --> DOS["DOS<br/>delegate only"]
  DOS -->| "DelegationPolicy" | H["Typed Handoff"]
  H --> OBS["OBS<br/>read-only brief"]
  H --> BOPA["BOPA<br/>prepare/verify/truth"]
  OBS --> Brief["OperationalBrief"]
  BOPA --> Child["AgentRun child"]
  DOS --> Audit["Append-only OrchestrationRun audit"]
```

25.4 Direct BOPA vs DOS+BOPA+OBS

```
Direct:  Owner → BOPA → services → stop
Multi:   Owner → DOS → (OBS | BOPA) → stop
Prefer:  Direct for ordinary prep
Use multi: learning, briefing, future specialists
```

25.5 Authority matrix

See §10.

25.6 Typed handoff lifecycle

```
stateDiagram-v2
    [*] --> pending
    pending --> policy_blocked: deny
    pending --> accepted: allow
    accepted --> executing
    executing --> completed
    executing --> stopped
    pending --> rejected
    pending --> stale
    pending --> cancelled
```

25.7 Happy-path sequence (`prepare_then_brief`)

```
sequenceDiagram
    participant O as Owner
    participant D as DOS
    participant B as BOPA
    participant S as OBS
    O->>D: run --goal prepare_then_brief --approve
    D->>D: observe SoT
    D->>B: handoff prepare
    B-->>D: AgentRun awaiting_owner
    D->>D: re-observe
    D->>S: handoff brief
    S-->>D: OperationalBrief
    D-->>O: stop + owner action
```

25.8 Truth-blocked sequence

```
sequenceDiagram
    participant O as Owner
    participant D as DOS
    participant S as OBS
    O->>D: run --goal brief --approve
    D->>D: observe truth=fail
    D->>S: handoff brief
    S-->>D: blockers + owner_remediate_truth
    Note over D,S: No waive / rewrite / submit
    D-->>O: briefing_complete
```

25.9 Rejected multi-agent theatre

See §6.

25.10 Major trade-offs

Trade-off	Choice
Product value vs learning	Learning proof only
Supervisor power	Delegate-only DOS
Chat vs typed handoffs	Typed + audited
LLM vs deterministic	Deterministic default
Broaden BOPA vs OBS	OBS for brief-only delta
Framework now	Custom contracts (ADR-003)

25.11 Five key lessons

1. Distinct permissions justify agents; renamed roles do not.
2. Supervisors must not become domain super-users.
3. Typed handoffs enable audit and fail-closed control.
4. Deterministic policy remains the admission authority.
5. Architecture success $\neq$ product default.

25.12 Ten interview questions

1. Why reject Prep/Truth/Review agent splitting?
2. What can DOS never do, and why?
3. How do DelegationPolicy and ToolPolicy differ?
4. How does a typed handoff prevent privilege escalation?

5. When should OBS run instead of BOPA?
6. How do you prevent circular specialist loops?
7. How does resume avoid duplicate preparation?
8. Why keep deterministic routing as default?
9. When would multi-agent become commercially stronger?
10. How do you explain FR-016 without overselling product value?

25.13 Concise model-answer points

1. Theatre = same tools, new names; need distinct allow-lists.
2. DOS: observe/select/hand off only — no prepare/submit.
3. Delegation admits specialists; ToolPolicy admits actions.
4. Handoff carries goal/hash/output kind — not tool grants.
5. Interviewing/truth/brief goals → OBS first.
6. Max steps, visit limits, repeated/circular denies.
7. Re-observe SoT; skip completed child / unchanged brief.
8. LLM never expands allow-lists; policy is sole admission.
9. Job Discovery / second mutating authority boundary.
10. Learning proof + substrate; prefer `cic agent run` daily.

25.14 Glossary

Term	Meaning in FR-016
Supervisor	DOS — deterministic coordinator that only delegates
Specialist	BOPA or OBS with its own ToolPolicy
Delegation	Admission of a specialist invocation by DelegationPolicy
Handoff	Typed, append-only request from supervisor to specialist
ToolPolicy	Per-specialist admission of actions
Authority boundary	What a specialist may / must not do
Source of truth	Opportunity / package / truth / pipeline domain records
Checkpoint	Recovery pointer on OrchestrationRun (not SoT)
Orchestration audit	Append-only parent events + handoffs + child refs
Privilege escalation	Gaining another specialist's tools via handoff or role

26. Tests

Suite	Result
tests/unit/multi_agent/ (incl. 20-case corpus)	PASS
tests/unit/agent/test_cli.py	PASS
scripts/run_fr016_m4_manual.py	PASS
Live interviewing brief	PASS (OBS only)

27. Definition of Done

Criterion	Status
M0 architecture accepted	Done
ADR-008 accepted	Done
M1–M4 delivered	Done
DOS / BOPA adapter / OBS	Done
Delegation + ToolPolicy separation	Done
Typed handoffs + audit + resume	Done
Owner CLI	Done
Corpus 20/20 + manual validation	Done
Honest product / learning / commercial assessment	Done
Study-aid source captured	Done
Documentation frozen + navigation consistent	Done
No FR-017 implementation	Confirmed
No scope leakage	Confirmed
Engineering Learning Academy ready	Confirmed

28. Acceptance recommendation

ACCEPT AND FREEZE — LEARNING PROOF COMPLETE

29. Final repository status

Item	Status
FR-016	Complete / Frozen / Accepted
Learning Proof	Complete
Engineering Learning Academy	Ready
M2 verdict	GO AS LEARNING PROOF ONLY (preserved)
Daily prep	<code>cic agent run</code>
Orchestration CLI	Optional learning/substrate
FR-017	Complete / Frozen (fr017_agent_evaluation_observability.md)
Horizon 1B	Not blocked on FR-017; start on owner request (usable 1A application loop)